EXPERIMENT 8 - LAB RECORD

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Permissions, roles, role hierarchy and data sharing for the Appointment_Booking_Form application

Submitted by	PRANAV SRINEVAS V
Register number	24101068
Application	Appointment_Booking_Form (Zoho Creator)
Platform	Zoho Creator (low-code)

1. Aim

To configure user management and role-based access control (RBAC) in the Appointment_Booking_Form application built in Zoho Creator: review the available permission sets, build a role hierarchy, and create a custom data-sharing rule that gives the Medical Administrator role Read/Write access to Appointment_Booking_Form records owned by the Receptionist role.

Objectives

- Review the application dashboard (Hospital_Appointment_Dashboard) and the appointment report.
- Study the permission sets (Read and Write) listed under User Permissions.
- Build the role hierarchy CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist}.
- Create the share rule Receptionist_to_Medical_Administrator on the Appointment_Booking_Form with Read/Write access.
- Understand how record ownership differs from record visibility.

Software and tools

- Zoho Creator (web-based low-code platform) with the Appointment_Booking_Form application
- Web browser with internet access
- Zoho account with permission to open the application in Edit mode

2. Theory

User management and RBAC

User management decides who may open an application and what each person can see or change. In Zoho Creator, users are linked to **permission sets** (profiles) and **roles**. Under *Settings* → *User Permissions* these controls appear on three tabs: Permissions, Roles and Data Sharing. RBAC assigns rights to roles rather than to individuals, which makes administration simpler, supports least privilege and keeps the access policy consistent and easy to audit.

Concept	Meaning
Users	People added to the application who sign in to work with its forms, reports and data.
Permissions	Named sets that define what a user may do with application components (for example Read and Write).

24101068

Concept	Meaning
Roles	Positions in the organisation tree; they decide where a user sits in the hierarchy and which records are visible.
Role hierarchy	A tree in which each role reports to another. A higher role can normally view records of the roles below it, not the reverse.
Data sharing	Share rules that extend visibility beyond the hierarchy, between roles or users, at Read Only or Read/Write level.

Application specification

Parameter	Detail
Application	Appointment_Booking_Form
Dashboard / report	Hospital_Appointment_Dashboard with the appointment_report
Permission sets	Read, Write
Role hierarchy	CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist}
Custom share rule	Receptionist → Medical Administrator on Appointment_Booking_Form, Read/Write

3. Procedure

1. Sign in and open the application — Sign in to Zoho Creator. The home page lists the available applications; open Appointment_Booking_Form.

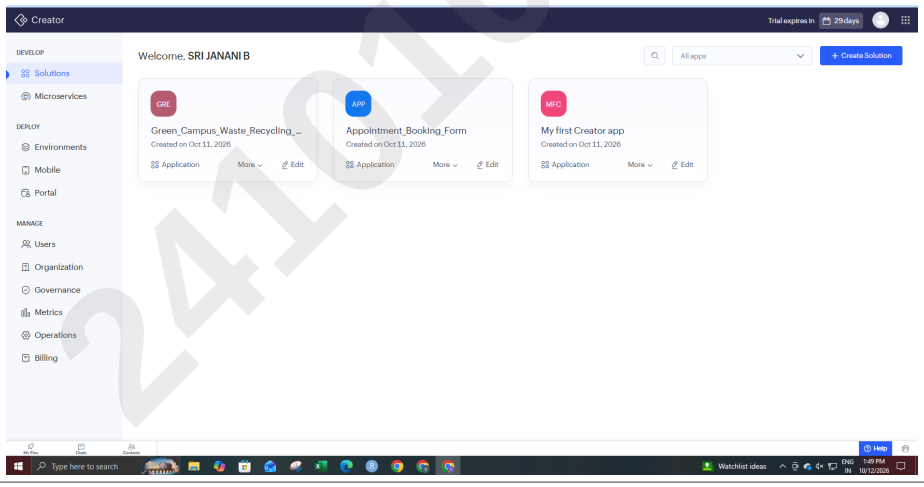


Figure 8.1: Zoho Creator home page

2. Review the dashboard — Open the application in Access mode. The Hospital_Appointment_Dashboard shows the age chart by department and the appointment_report grouped by department.

24101068

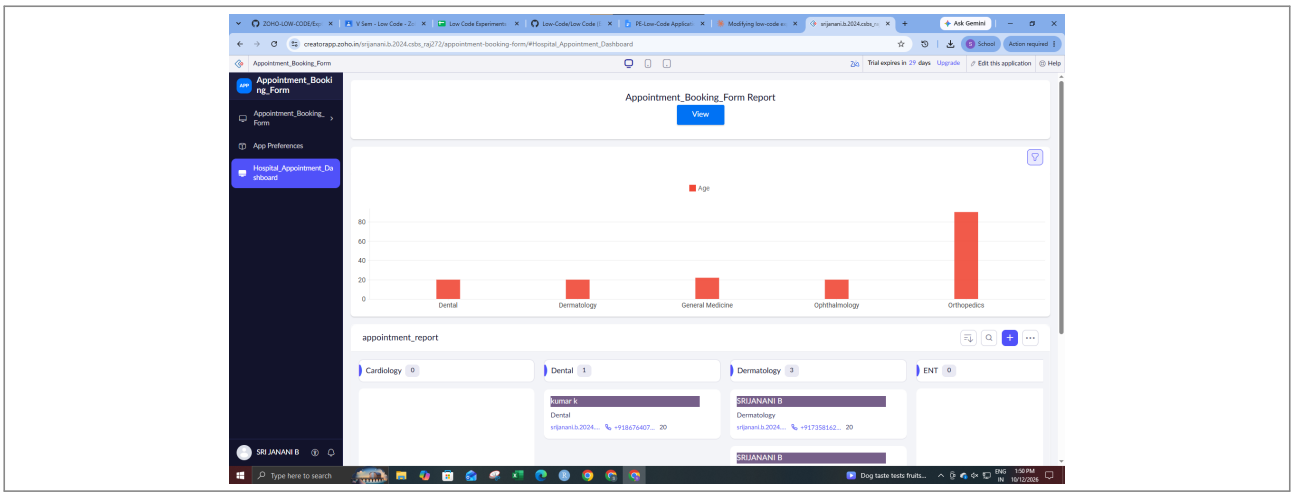


Figure 8.2: Hospital_Appointment_Dashboard

3. Enter Edit mode and open Settings — Choose *Edit this application*, then select the *Settings* tab from the top navigation bar.

4. Open User Permissions — Under the Permissions group select *User Permissions*. The Users page opens with the Permissions, Roles and Data Sharing tabs. The Permissions tab lists the Read and Write permission sets.

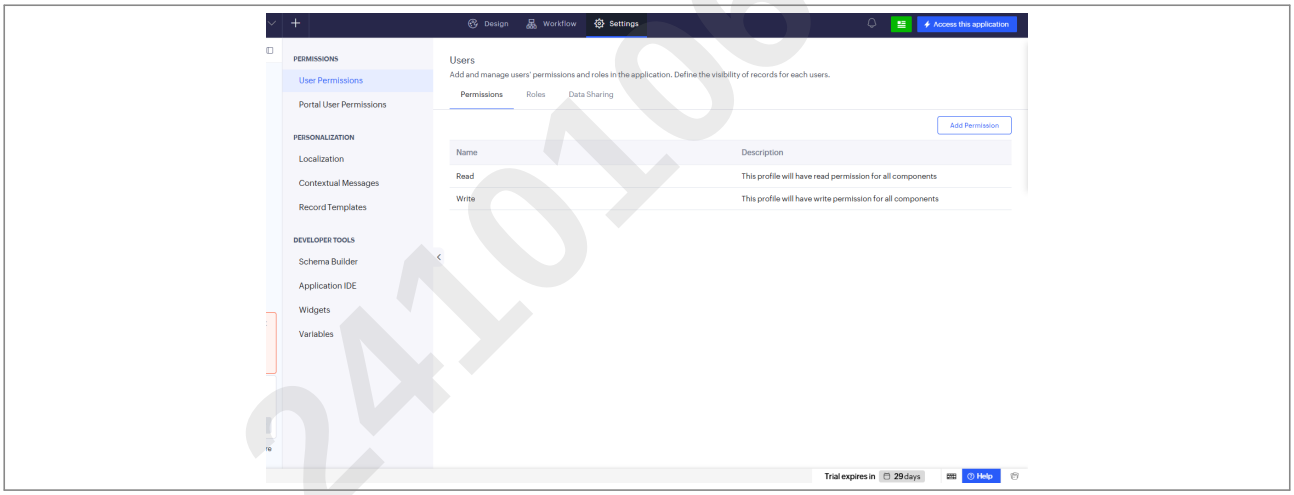


Figure 8.3: Permissions tab

5. Build the role hierarchy — Open the Roles tab and use *Add Role* / *Manage Role Hierarchy* to arrange: CEO at the top, Medical Administrator beneath it, and Head of Department, Department Coordinator and Receptionist beneath Medical Administrator.

2410106

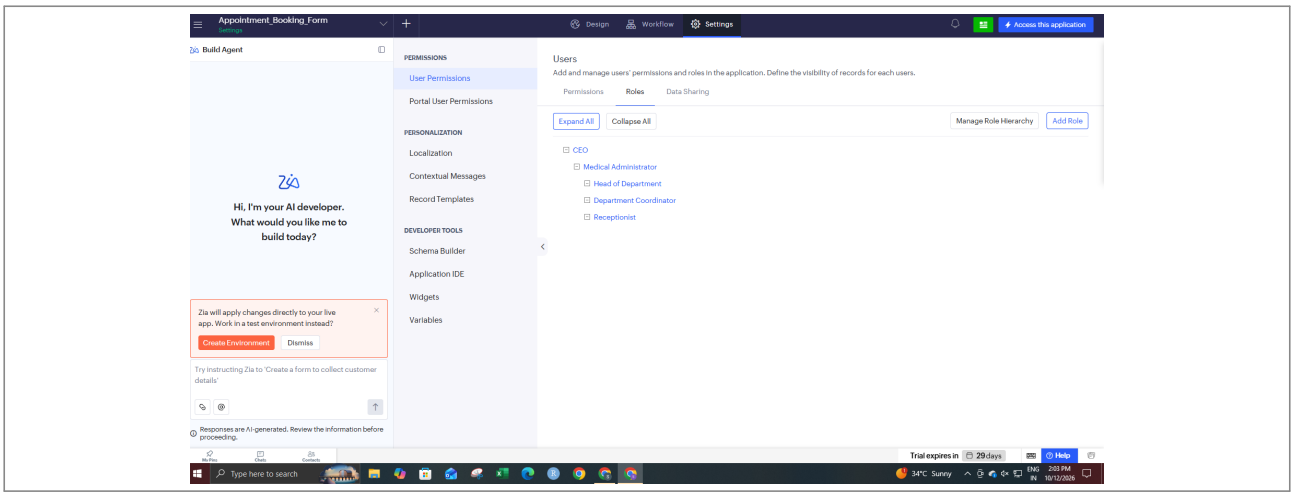


Figure 8.4: Roles and role hierarchy

6. Create the share rule — Open the Data Sharing tab and choose *New Share Rule*. Name it Receptionist_to_Medical_Administrator, set the form to Appointment_Booking_Form, records shared from Roles → Receptionist, records shared to Roles → Medical Administrator, and level of access to Read/Write.

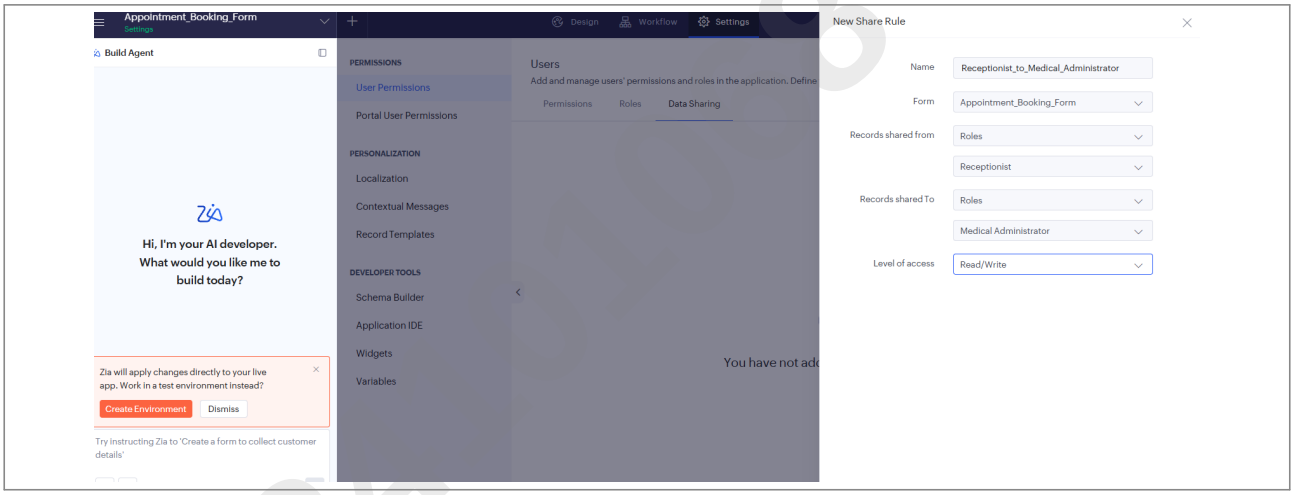


Figure 8.5: New Share Rule panel

7. Save and verify — Save the rule. Confirm it appears in the Data Sharing list, then reopen it with *Edit Sharing Rule* to check that every field was stored as configured.

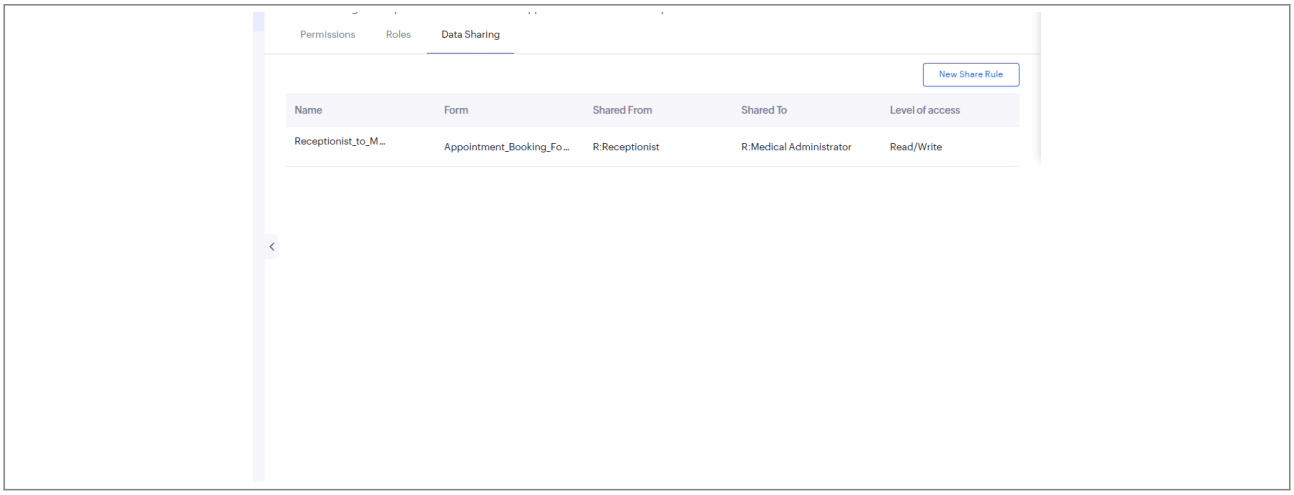


Figure 8.6: Saved rule in the Data Sharing list

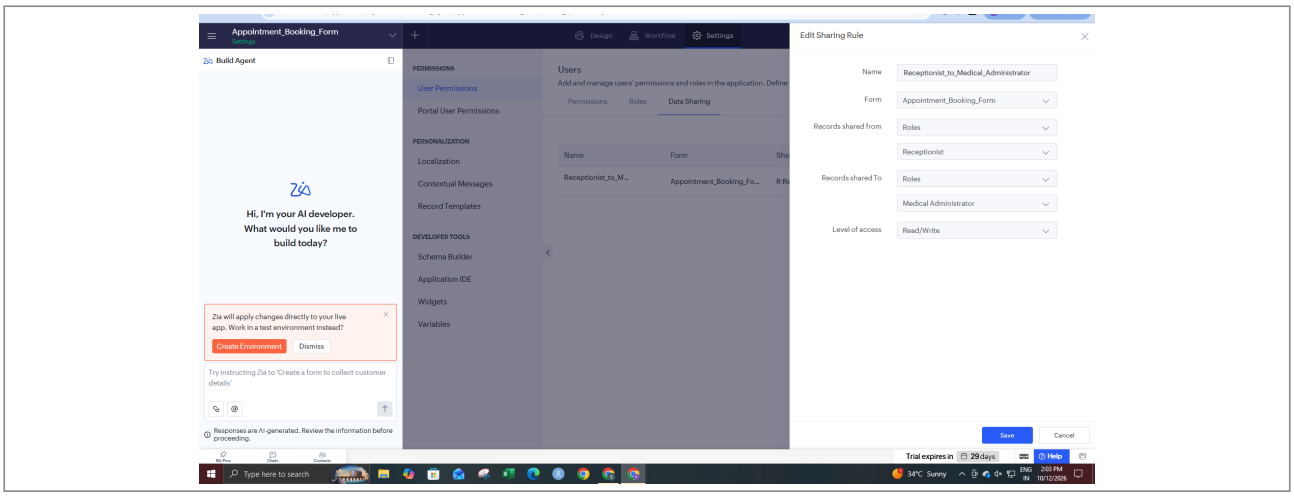


Figure 8.7: Edit Sharing Rule panel confirming the saved settings

4. Role hierarchy and data sharing

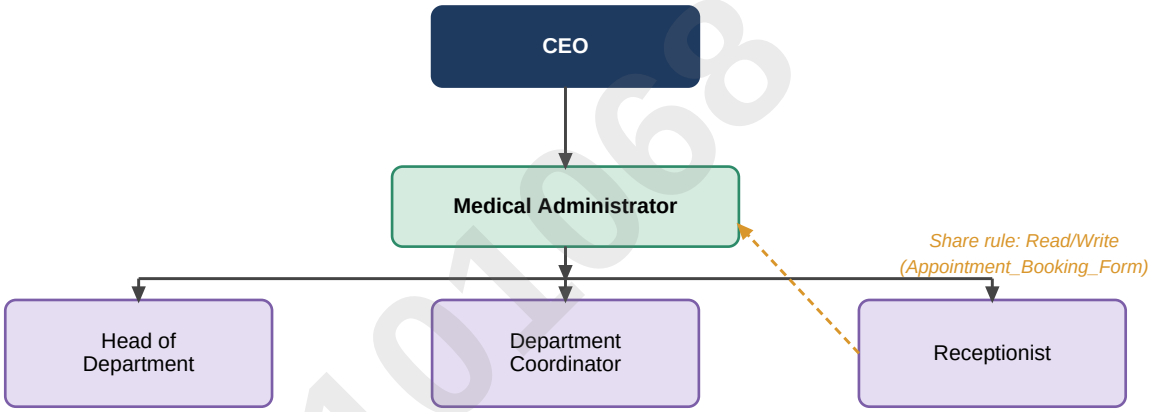


Fig. 7: Role hierarchy with the Receptionist → Medical Administrator share rule.

By default the hierarchy gives upward visibility only: a higher role sees the records of roles below it, but not the reverse. The share rule adds a deliberate, auditable Read/Write channel from the Receptionist role, which captures new patient profiles and appointment details, to the Medical Administrator, who monitors and updates appointment status (pending, confirmed, cancelled). This avoids granting the Medical Administrator system-wide Write permission.



Fig. 8: Flow of Appointment_Booking_Form record access through the share rule.

RBAC working

Level	Role	Intended function
1	CEO	Overall oversight; can view records of the roles reporting to it.
2	Medical Administrator	Monitors appointment statuses; gains Read/Write on Receptionist-owned appointment records through the share rule.
3	Head of Department, Department Coordinator, Receptionist	Day-to-day work: reviewing assigned cases, managing department masters and capturing new patient profiles.

5. Result

- The Appointment_Booking_Form application and its dashboard were reviewed.
- The Read and Write permission sets were studied under User Permissions.
- The role hierarchy CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist} was created.
- The share rule Receptionist_to_Medical_Administrator (Appointment_Booking_Form, Read/Write) was configured, saved and verified.

6. Conclusion

The experiment showed how permissions, roles, the role hierarchy and data sharing work together as a role-based access control mechanism in Zoho Creator. The three-level hierarchy and the directional Read/Write share rule keep information channels controlled: the Medical Administrator gets exactly the access needed to monitor and update appointment status, the CEO keeps overall oversight, and the Receptionist, Department Coordinator and Head of Department continue to own their respective records.

24101068